Updated Figures, Tables, and Abstract for Kim et al. 2026

A Note to the Reader

November 6, 2026

Citation

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Purpose

This document was created to show updated figures, tables, and abstract in our article as we continue to improve our PE deal list. These figures, tables, and abstract reflect the PE hospital acquisition dataset as available on the November 6, 2026. This document will be updated following updates to the PE acquisition dataset. A full tracker of changes to the PE acquisition dataset can be found in the README.

Disclaimers

Disclaimer 1: GitHub Availability
The data uses for these updated figures, tables, and abstract, along with a complete tracker of all updates to this data can be found on our GitHub. This document can also be found on our GitHub: https://github.com/sungilkim94/PE-Healthcare.

Disclaimer 2: Paper Availability
The orginal publication of our article is available through *Health Affairs Scholar*. This document presents updates from the figures, tables, and abstract shown in that paper based on improvements to our PE hospital acquisition dataset. The orginal paper can be found at:
https://academic.oup.com/healthaffairsscholar/article/4/4/qxag071/8626085.

A Practitioner's Guide to Using Data on Private Equity Hospital Acquisitions

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Abstract

Introduction: Private equity (PE) investment in US hospitals has attracted substantial policy and research attention, but empirical work has been limited by fragmented and inconsistent transaction data. We aimed to construct a more comprehensive, validated dataset of PE ownership of US hospitals and to provide a practical guide for using these data in research.

Methods: We integrated six major commercial deal databases to identify PE investments in US hospitals from 2000 to 2024. We filtered transactions to PE-related hospital deals, matched targets to AHA and CMS hospital identifiers, manually verified uncertain matches, reconciled duplicate transactions across sources, expanded system-level deals to constituent hospitals, and verified deal and exit dates.

Results: We identified 138 unique PE deals involving 539 unique short-term acute care hospitals, corresponding to 704 hospital-deal observations. The six databases differed substantially in deal coverage, deal type, and whether transactions were reported at the hospital or system level. Reliance on a single source would therefore omit many valid deals and can produce biased or incomplete analytic samples. We also found that linking transactions to stable hospital identifiers required substantial manual verification because of system-level transactions, inconsistent reporting, and identifier changes over time.

Conclusion: Accurate study of PE ownership in hospitals requires multi-source data construction, transparent validation, and careful linkage to stable hospital identifiers. This

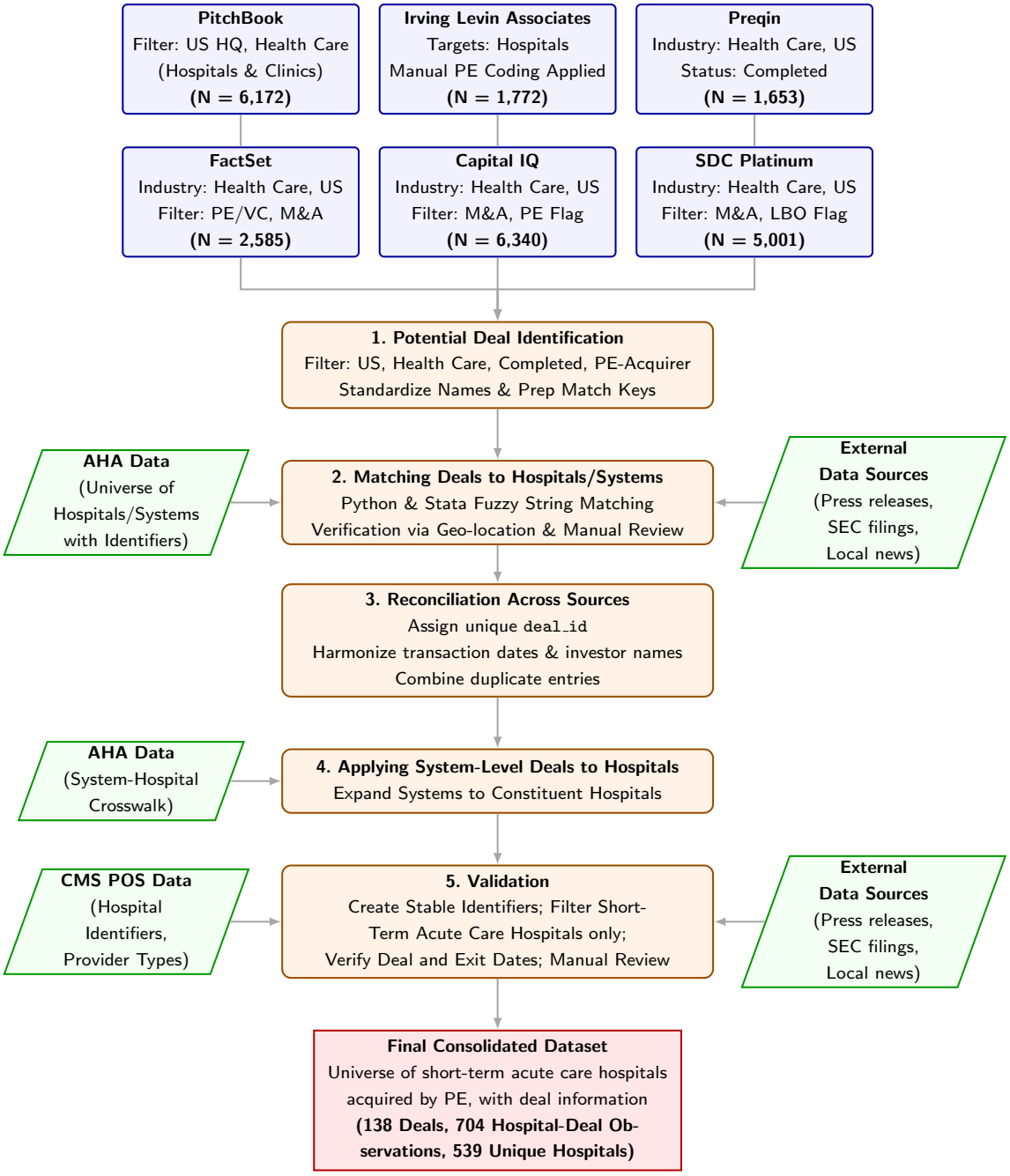
*The author ordering reflects Kim's primary role as a "first author" and Naslund and Wang's significant contributions as "second authors." The remaining authors are listed alphabetically. sungil.kim@duke.edu.

†Conflict of Interest Disclosures: McDevitt has received funding from the American Investment Council, Duke University, and Arnold Ventures; personal fees from the American Society of Nephrology, Health Management Academy, and Renalogic; and payments as a consultant and litigation expert for Charles River Associates on matters in the health care sector. mcdevitt@wustl.edu.

harmonized dataset and workflow provide infrastructure for more accurate, transparent, and replicable research on PE ownership in the US hospital sector.

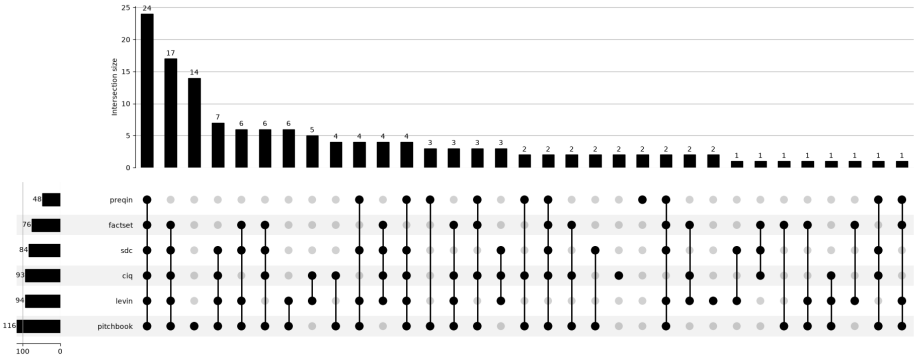
Keywords: Private Equity; Hospital Acquisitions; Data Linkage; Reproducibility

Figure 1: Data Construction Flowchart

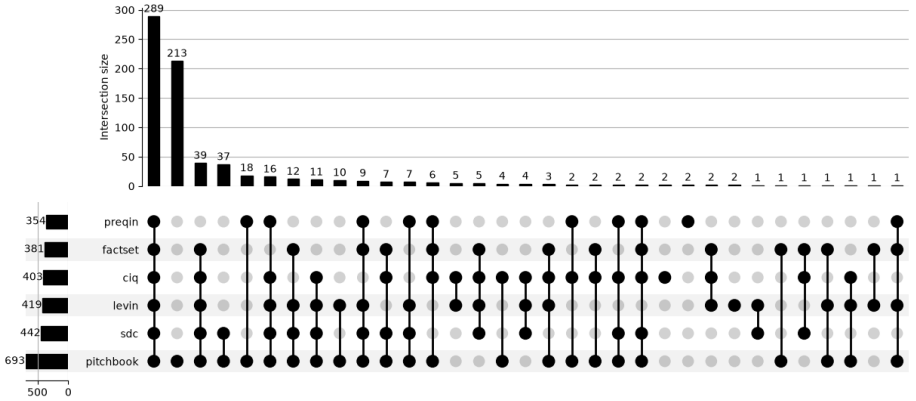


Note: This flowchart summarizes the overall data construction workflow used to assemble the hospital-level private equity acquisition dataset. It is intended as a visual guide to the process described in Section 2 and Appendix A.

Figure 2: Overlap of Data Sources



(a) Deals



(b) Hospitals

Note: This figure presents UpSet plots summarizing overlaps among data sources used to construct the PE hospital acquisition sample from 2000–2024. Panel A (Deals) reports intersections at the deal level, where each observation corresponds to a unique transaction. Panel B (Hospitals) reports intersections at the hospital level, where each observation corresponds to an individual hospital involved in a transaction. Bars report the number of deals or hospitals associated with each unique combination of sources, while connected dots indicate which sources contribute to each intersection. Counts are aggregated over the full sample period.

Table 1: Comparison of Commercial Databases

	PitchBook	Levin	Preqin	FactSet	SDC	Capital IQ	All Sources
Panel A: An Observation is a Unique Hospital							
Unique Hospital Count	528	367	333	348	390	361	539
Panel B: An Observation is a Hospital-Deal Pair							
Hospital-Deal Count	678	421	355	382	444	402	704
Deal Level							
System Level	86.6%	78.5%	90.4%	82.9%	82.6%	80.1%	84.1%
Hospital Level	13.4%	21.5%	9.6%	17.1%	17.4%	19.9%	15.9%
Deal Type							
LBO	72.7%	98.3%	98.9%	99.2%	91.0%	98.5%	73.6%
PIPE	21.1%	0.0%	0.0%	0.3%	6.5%	0.3%	20.3%
Growth Equity	6.2%	1.7%	0.5%	0.7%	2.5%	1.2%	6.1%
Hospital Characteristics							
For-Profit	75.0%	69.9%	76.3%	71.6%	72.8%	72.5%	73.6%
Rural	34.6%	31.59%	33.3%	34.8%	35.8%	29.5%	34.1%
Bed Count	197	216	217	223	211	216	197
Panel C: An Observation is a Deal							
Deal Count	114	94	48	77	85	92	138
Deal Level							
System Level	37.0%	26.1%	45.6%	32.19%	33.3%	30.0%	32.8%
Hospital Level	63.0%	73.9%	54.4%	67.1%	66.7%	70.0%	67.3%
Deal Type							
LBO	85.1%	96.8%	93.8%	96.1%	95.3%	94.6%	87.0%
PIPE	4.4%	0.0%	0.0%	1.3%	1.2%	1.1%	3.6%
Growth Equity	10.5%	3.2%	6.2%	2.6%	3.5%	4.3%	9.4%

Note: Panel A reports unique hospital counts across the commercial databases. Panel B reports characteristics at the "hospital-deal" level. This includes hospitals that have been targets in a PE deal more than once. Panel C reports characteristics at the deal, rather than hospital, level. "System Level" deals include only deals involving the full hospital system. "Hospital Level" deals include single-hospital and multi-hospital acquisitions by a single buyer (e.g., five Miami HCA Hospitals). For-Profit: the share of hospitals with for-profit status in the year of acquisition as recorded in the CMS POS file. Rural: the share of hospitals with rural status in the year of acquisition as recorded in the CMS POS file (Note that urban/rural attributes are not available prior to 2011, so only hospitals with deal years 2011-2024 are included in these shares). Bed Count: average total bed count among hospitals as recorded in the CMS POS file in the year of acquisition.

Table 2: Variation in Deal Sources and Counts Across the Literature

	Hospital Count*			Sources Used								
Study	**Original**	**Expanded**	**Maximum**	**PB**	**Levin**	**Preqin**	**CIQ**	**FS**	**SDC**	**Zephyr**	**CBI**	**PESP****
Bhatla et al. (2025)[1]	73	104	186	✓	✓							✓
Bruch, Gondi, and Song (2020)[2]	204	248	379	✓	✓		✓					
Bruch, Zeltzer, and Song (2021)[3]	130	134	134[a]		✓							
Cerullo et al. (2021)[4]	228	239	278	✓						✓	✓	
Cerullo et al. (2022)[5]	184	234	257	✓						✓	✓	
Cerullo et al. (2022)[6]	257	281	342	✓						✓	✓	
Diaz et al. (2025)[7]	67	123	323									✓
Gao, Kim, and Sevilir (2025)[8]	419[b]	357	634	✓		✓	✓	✓	✓			
Kannan, Bruch, and Song (2023)[9]	51	127	186	✓	✓							
Kannan et al. (2025)[10]	49	113	186	✓	✓		✓					
Liu (2022)[11]	838[b]	352	561	✓	✓	✓	✓		✓			
Offodile et al. (2021)[12]	282	286	430	✓						✓	✓	
This paper			**704**	✓	✓	✓	✓	✓	✓			

Note: This table compares the counts of hospitals with PE investment across previous studies in the literature. PB = PitchBook; CIQ = Capital IQ; FS = FactSet; SDC = SDC Platinum; CBI = CB Insights.

*The "Original" column reflects the reported hospital count in each study. Because these counts are in part determined by analytic sample restrictions, we report the unique hospital counts calculated by applying these restrictions (described in Table **??**) on our dataset in the "Expanded" column. This "Expanded" column should therefore approximate each study's sample using our dataset. We also report the total possible PE hospital count over the relevant period considered by each study in the "Maximum" column. The values reported in the "Maximum" column are hospital-deal observations rather than unique counts.

**The sources in the right-most panel (Zephyr, CBI, PESP) were reviewed but excluded from our primary data construction because they did not provide unique deals beyond our existing primary sources. PESP (Private Equity Stakeholder Project) is a tracker of current hospital ownership that compiles lists from public sources rather than a primary transaction database. SEC filings were sometimes used for deal verification but are excluded here as they do not constitute a formal transaction database. Our deal list completely subsumes deals reported in Zephyr and CBI in the relevant time periods.

[a] Bruch, Zeltzer, and Song (2021)[3] limit their sample to active PE investments. We use our verified exit dates to determine if PE investments were active in 2018. Exit dates are only tracked for LBOs. PIPE and growth equity deals are not included in these counts.

[b] Gao, Kim, and Sevilir (2025)[8] restrict to AHA-defined general medical and surgical hospitals, while Liu (2022)[11] includes all CMS hospital types (including critical access hospitals, long-term care hospitals, rehabilitation hospitals, and psychiatric hospitals). Our sample instead focuses on CMS-defined short-term acute care hospitals.